PORTFOLIO · AUDIO ENGINEERING · DSP · SPATIAL AUDIO

Kritan Duwal

Audio engineer who writes the software.

M.S. Audio Engineering on top of a B.S. in Computer Science and three years as a software engineer. I measure rooms, build DSP and web-audio tools, and record and mix music.

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LinkedIn GitHub churchauralheritage.org AES paper All audio, code and reports

M.S. Audio Engineering, GPA 4.0 · Belmont University, Nashville, TN · Aug 2024 - May 2026
B.S. Computer Science, cum laude (GPA 3.64) · University of Texas at Arlington · Aug 2018 - May 2021

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01 Aural Heritage web app
02 AES 159th Convention paper
03 Hybrid Pan JUCE plugin (C++)
04 MATLAB DSP effects
05 IR-based system modeling
06 Audio Precision measurement
07 Samba Funk (AES Student Recording Competition)
08 Mixing coursework (technique studies)
09 Mastering, tape transfer, restoration
10 Guidance-based generative audio
11 Software Engineer, First Command

01 RESEARCH & SPATIAL AUDIO · JUN 2025 - PRESENT

Aural Heritage: hear a historic church from where you stand

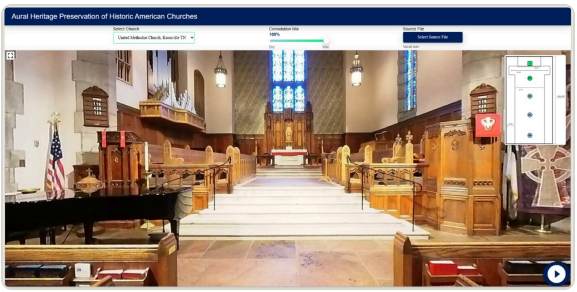
Research Assistant, Belmont University (Dr. Doyuen Ko)

I measured impulse responses in historic US churches, then built churchauralheritage.org: pick a church, click a measured listening position inside its 360° photo, and any audio plays back in the real acoustics of that spot.

- On-site measurement with Room EQ Wizard, omnidirectional and first-order ambisonic microphones, and calibrated loudspeaker excitation at multiple positions per room.
- Real-time convolution in the browser with the Web Audio API. Nothing is simulated: you hear the measured building.
- Head-tracked binaural pipeline: A-format ambisonic IRs are converted to B-format and decoded to binaural with HRTFs; the sound field rotates with the 360° view.
- Binaural and stereo renders are loudness-matched with an ITU-R BS.1770 LUFS measurement.
- Plain HTML/CSS/JavaScript with no build step, deployed on Netlify, with a Node test suite covering the audio graph, room data and UI.

`JavaScript` `Web Audio API` `Ambisonics` `HRTF / binaural` `ITU-R BS.1770` `Room EQ Wizard` `Node test runner` `Netlify`

Open the live app Listen to the audio



The web app: a 360° view with clickable measured positions.

02 RESEARCH & SPATIAL AUDIO · APR 2025 - OCT 2025

Two is Better Than One: time- and level-based panning

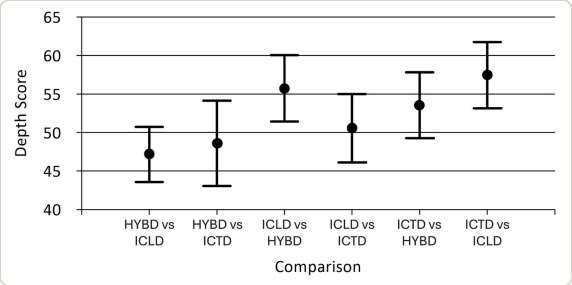
Co-authored with Austin Arias, Dr. Wesley A. Bulla and Dr. Song Hui Chon

A listening study on whether combining inter-channel time difference (ICTD) with level difference (ICLD) improves stereo panning. Presented as a peer-reviewed Express Paper at the 159th AES Convention, Long Beach, October 2025.

- Thirty stimuli built from ten multitrack recordings, panned with ICLD, ICTD or both (HYBD).
- Listeners rated width, depth, externalization and envelopment in counterbalanced pairs, on headphones and on loudspeakers.
- On headphones, depth was rated significantly higher for the hybrid technique.
- On loudspeakers, listeners strongly preferred the hybrid and preferred against level-only panning.

`Experimental design` `Listening tests` `Statistics` `Spatial audio` `Technical writing`

Paper in the AES E-Library



Mean depth scores per comparison (headphone condition).

03 DSP & PLUGINS · 2026

Hybrid Pan: the paper, as a real-time plugin

Developer

A C++17 / JUCE port of the three panning laws from the AES paper, built as VST3, AU and Standalone. Try the three laws in the live demo.

- ICLD is a constant-power level pan; ICTD delays the far channel by a sub-sample amount; HYBD combines them, across the whole band or split at a crossover (time below, level above).
- Streaming third-order Lagrange fractional delay on a ring buffer. The offline MATLAB reads one sample ahead, so the plugin runs a constant one-sample latency and reports it to the host for delay compensation.
- Second-order Butterworth crossover (juce::dsp::IIR) with both bands always computed, so mode switches are click-free.
- Parameters through APVTS with 20 ms smoothing. Panning math and interpolator verified numerically against the MATLAB reference.

Also in progress: a JUCE multi-effects plugin (phaser, chorus, overdrive, ladder filter) with a user-reorderable chain passed to the audio thread through a lock-free FIFO.

`C++17` `JUCE` `CMake` `VST3 / AU` `Fractional delay` `IIR filters` `MATLAB`

GitHub

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04 DSP & PLUGINS · AUG 2025 - MAY 2026

DSP effects and synthesis, written from the difference equation up

Graduate coursework (Digital Audio, Audio Signal Processing)

Effects, oscillators and analysis tools implemented in MATLAB without toolbox shortcuts. Each one ships with a test script and before/after audio.

- Transistor-style clipper with memory: the threshold sags the longer the signal stays in saturation, giving a sloped clip rather than a flat one.
- Mid-side distortion (hard clip on the mid channel only, so a drum bus keeps its width), three-path parallel distortion, and an asymmetric waveshaper.
- Aural exciter, stereo echo with feedback and square-law panning, wavetable and phase-accumulator synthesis, ring modulation, a self-oscillating IIR sine oscillator.
- Pitch detection by autocorrelation and zero-crossing rate; a hand-written DFT, inverse DFT and spectrogram.
- Ambisonics: A-format to B-format encoding from tetrahedral IRs, and B-format sound-field rotation for head tracking.

`MATLAB` `Nonlinear processing` `FIR / IIR design` `Synthesis` `DFT` `Ambisonics`

Listen to the audio

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05 DSP & PLUGINS · SPRING 2026

Can an impulse response stand in for the hardware?

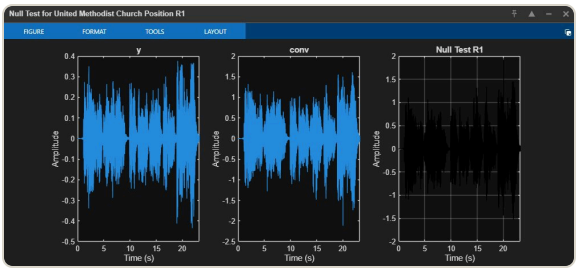
Team project with Austin Arias and Sihyeon Park

We measured plugins, a GML 8200 hardware EQ and a church, modeled each from its impulse response, and null-tested the model against the real thing.

- Plugins: captured the IRs of a Pro Tools EQ and delay, then compared convolution against the plugin output.
- Hardware: measured a GML 8200 parametric EQ with a sine sweep, estimated its transfer function by spectral division and Tikhonov-regularized deconvolution, and aligned by cross-correlation.
- Room: applied the same method to an impulse response measured in a church.

`MATLAB` `Deconvolution` `Tikhonov regularization` `Cross-correlation` `Null testing` `Pro Tools`

Listen to the audio



Null test: measured output, convolved model, and the residual.

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06 MEASUREMENT & ACOUSTICS · APR 2025

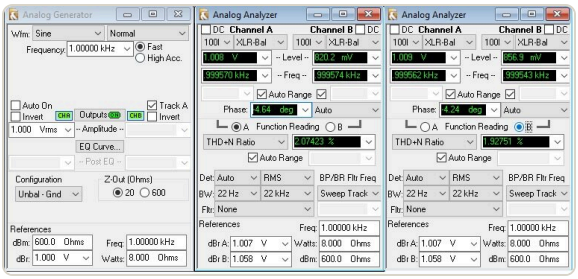
Bench-testing audio interfaces on an Audio Precision AP2700

Acoustics coursework (Dr. Doyuen Ko)

Measured two Focusrite Scarlett interfaces (2i2 and 2i4) and compared the results with the published specifications.

- Frequency response, THD+N, inter-channel phase at 100 Hz, 1 kHz and 10 kHz, crosstalk and signal-to-noise ratio.
- The 2i4 measured 0.0032% THD+N; crosstalk at 1 kHz was -80.5 and -81.7 dB.

`Audio Precision AP2700` `THD+N` `Crosstalk` `Frequency response` `Technical reporting`



Frequency response sweep, both channels.

07 RECORDING AND MIXING · RECORDED OCT 2024, MIXED FEB - APR 2026

Samba Funk: a session I recorded, edited and mixed

Recording, editing and mix engineer (supervised by Dr. Wesley A. Bulla)

The Belmont School of Music Faculty Jazz Quartet, tracked on an SSL Origin in Robert E. Mulloy Studio A at 48 kHz / 24-bit. Submitted to the AES Student Recording Competition, Nashville 2026 (Category 2: Studio Recording). I also made a Dolby Atmos 7.1.2 mix.

- Glyn Johns overheads (two U87s equidistant from the snare), FET U47 on kick, C12 VR on acoustic bass, three-mic grand piano, guitar on DI plus an isolated SM57 amp.
- Clean, direct capture with no processing on the way in, to leave every tonal decision for the mix.
- Mix: notch EQ around 500 Hz to clear low-mid build-up, gentle compression on kick and bass, Chow Tape Model on the mix bus, and the guitar run through a hardware Teletronix LA-2A.
- The lesson I kept: I first edited all the bleed out of the drums, then put it back. For this vintage jazz feel, a little spill is part of the sound.

`SSL Origin` `Pro Tools` `Microphone technique` `Teletronix LA-2A` `Tape saturation` `Dolby Atmos 7.1.2`

Listen to the audio



Robert E. Mulloy Studio A, from the control room.

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08 MIXING COURSEWORK · SPRING 2026

Mixing coursework: one technique at a time

Graduate mixing course (AET 5895)

Technique studies from my mixing course. Each one applies a single technique to a 30-second section of a multitrack, bounced raw and processed, so you can hear exactly what that one decision did. These are exercises, separate from the Samba Funk session above.

- Each example is documented in a written mix log: session edits, fader and pan positions, and plugin settings.
- Covers level balance, EQ, compression, saturation and sample layering.

`Pro Tools` `Compression` `EQ` `Saturation` `Sample reinforcement` `Balance`

Listen to the audio

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09 MASTERING · SPRING 2025

Mastering: two Nashville rock tracks, plus tape and restoration

Mastering engineer (Audio Mastering and Applied Critical Listening)

Mastered "Nowhere to Go" (Velvet Mirage) and "Shudder" (Sour Tooth) in Steinberg WaveLab over several feedback rounds, and delivered main, instrumental, a cappella and TV versions as WAV, FLAC and AAC.

- Chain: resonance control and tonal EQ (Pro-Q3), multiband dynamics (Pro-MB), harmonic excitation (Ozone Exciter), limiting with true-peak control (Ozone Maximizer); outboard NSEQ-2 through the patch bay in one round.
- Analog transfer: 30 ips two-track stereo tape on a Studer A80 to 96 kHz / 24-bit.
- Restoration with iZotope RX: de-hum, spectral de-noise and de-click on archival recordings.

`Steinberg WaveLab` `FabFilter Pro-Q3 / Pro-MB` `iZotope Ozone` `iZotope RX` `Studer A80` `Loudness normalization`

Listen to the audio

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10 MACHINE LEARNING · AUG 2025 - DEC 2025

Steering a generative audio model with musical guidance

Team of 15 (Dr. Scott Hawley, Deep Learning and AI Ethics)

A flow model for audio generation built on Stable Audio Open, an open-source text-to-audio latent diffusion model, with guidance that targets chord, melody and beat.

- Implemented guidance mechanisms that steer generation toward musically coherent output.
- Separately, for the class leaderboard: a ResNet convolutional VAE with a 4×4 spatial latent, plus a flow-matching model trained in that latent space (PyTorch).

`PyTorch` `Latent diffusion` `Flow matching` `VAE` `Hugging Face` `Weights & Biases`

Project write-up

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11 SOFTWARE ENGINEERING · JUL 2021 - JUN 2024

Three years shipping financial software

Software Engineer, First Command Financial Services, Fort Worth, TX

Event-driven back-end services, a planning web app, and large data migrations, in an Agile release train.

- .NET Core Azure Functions with Azure Service Bus triggers for asynchronous, event-driven processing.
- An Angular (TypeScript) application that helps financial professionals plan and generate client recommendations.
- SQL to filter and migrate 600k+ records with data integrity intact.
- "Ninja Coder" award for leading testing that delivered zero defects in User Acceptance Testing.

`C# / .NET Core` `Azure Functions` `Azure Service Bus` `Angular` `TypeScript` `SQL` `Azure DevOps` `NUnit` `Selenium` `Terraform`

Experience and skills

EXPERIENCE

Research Assistant, Belmont University, Nashville, TN
June 2025 - present
Under the supervision of Dr. Doyuen Ko, took acoustic measurements to capture impulse responses (IRs) of historically significant churches in the US using Room EQ Wizard (REW), omnidirectional and ambisonic microphones, with calibrated loudspeaker excitation across multiple spatial positions.

Graduate Assistant, Belmont University, Nashville, TN
August 2025 - Dec 2025
Supported six graduate students in a Studio Systems and Operations course, providing hands-on instruction in recording techniques and operation of a Solid State Logic (SSL) console.

Data Analyst, Belmont University, Nashville, TN
May 2025 - August 2025
Delivered Power BI dashboards for the Provost and Deans while independently managing two student workers.

Software Engineer, First Command Financial Services Inc., Fort Worth, TX
July 2021 - June 2024
Designed and implemented .NET Core Azure Functions with Azure Service Bus triggers to handle asynchronous, event-driven processing, enabling reliable message handling and automated workflow execution.

ALSO

Arena room-acoustic measurement. A room-acoustic survey of a 245 × 215 × 58 ft arena: a Sabine reverberation estimate from surface materials, then measurements at seven positions to compare against it.

Studio sessions (Columbia Studio A, Mulloy Studio A). Ensemble sessions mixed through the console with Neve preamps and outboard gear: dbx 160 and 1176LN compressors, a plate reverb, a Lexicon reverb unit and a Yamaha SPX900 effects unit. A year later I was the Graduate Assistant for the course, giving hands-on instruction to six graduate students.

LifeFit (UTA senior design, State Farm). A proof-of-concept Android app exploring health-based insurance

SKILLS

Programming languages. MATLAB, Python, JavaScript, TypeScript (Angular), C#, Java, C++ (JUCE), C

Audio DSP. Filter design (FIR / IIR), fractional delay, nonlinear effects, convolution and deconvolution, ambisonics (A/B-format), HRTF binaural, LUFS (ITU-R BS.1770), Web Audio API

Measurement. Room EQ Wizard, impulse responses, sine sweeps, Audio Precision AP2700, RT60, null testing

Backend & API development. Spring Boot (REST & SOAP services), REST APIs, SOAP web services, object-oriented design, .NET Core, Azure Functions, Service Bus

Machine learning. PyTorch, Hugging Face, Weights & Biases, scikit-learn, XGBoost

DevOps & practices. Git, Azure DevOps, CI/CD pipelines, Terraform, Agile / SAFe Scrum (PI Planning), TDD, NUnit, Selenium

Database query languages. SQL, SOQL (Salesforce), KQL (log and telemetry analytics)

Recording. Close and stereo miking: ORTF, NOS, XY, AB spaced pairs, Blumlein; SSL consoles

Mixing. Hybrid analog/digital workflows; Avid Pro Tools editing, routing and automation; Dolby Atmos 7.1.2

Mastering. Steinberg WaveLab: EQ, multiband dynamics, limiting, loudness normalization, distribution-ready masters; iZotope RX restoration

Data. Power BI, ANOVA, experimental design, pandas, NumPy, R, Excel Solver (linear programming)

Systems programming. C, POSIX processes (fork / exec / wait), pthreads (mutexes, condition variables), FAT32, page replacement, ARM assembly, Make

Embedded. Teensy / Arduino (ADC, PWM), Raspberry Pi (Raspbian), C and assembly interop

Mobile and web. Android (Java), Firebase Authentication and Realtime Database, SQLite, Retrofit, PHP, HTML / CSS

Cloud and data stores. Azure (Functions, Service Bus), AWS S3 (boto3), MongoDB, Redis, Docker, Nginx (team project exposure)

quoting from Fitbit data, built over eight Scrum sprints and shown at UTA Innovation Day 2021.

Compilers and languages. flex, bison, grammars, ASTs

Computer vision and classical ML. MATLAB image processing, Hough transform, PCA, HOG, DTW; Naive Bayes, decision trees, k-NN, k-means, neural networks from scratch

Graphics and networking. 3D transforms, projection, clipping, Bezier surfaces; sockets, HTTP, distance-vector routing

Software design. OOP in C++ and Java, MVC, UML, requirements analysis, test plans, Scrum sprints

Soft skills. Critical listening, team collaboration, teaching, release cycle and rollback procedures, adaptability

Languages. English, Nepali, Hindi, Japanese